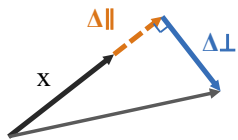


Geometric Decomposition



$$\Delta = \Delta_{\parallel} + \Delta_{\perp}$$

$$\Delta_{\parallel} = \alpha x$$

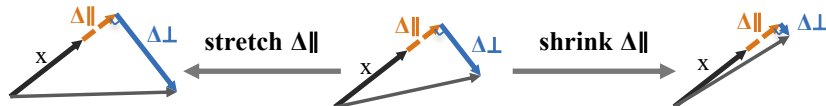
$$\Delta_{\perp} \perp x$$

Scaling-like
 $\Delta_{\parallel}$ change magnitude

Direction-changing
 $\Delta_{\perp}$ change direction

Manipulate the Perpendicular Component

Intervention: stretch or shrink $\Delta_{\perp}$ while keeping the rest of the update fixed



Changing $\Delta_{\perp}$ sharply hurts performance.

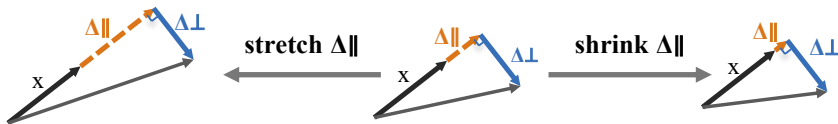


Performance



Manipulate the Parallel Component

Intervention: scale $\Delta_{\parallel}$ up or down while keeping the rest of the update fixed



Changing $\Delta_{\parallel}$ has limited impact on performance.

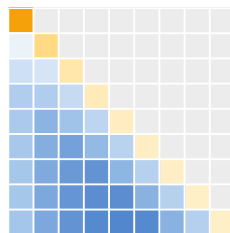


Performance

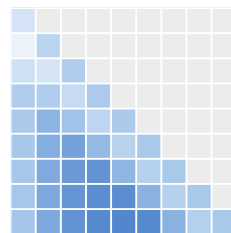


Attention-side View of Parallel Editing

Baseline Attention



Diagonal-suppressed



Diagonal Editing



Diagonal suppression redistributes mass off diagonal, linking the parallel edits.